# Rational Functions & Limits

Name: ____________

Date: ____________

## Student learning objectives for this unit

SUPPORTING SKILLS LEGEND

I — Introduce | D — Deepen | A — Apply | R — Reinforce

☐ **1. I can identify holes and vertical, horizontal, or slant asymptotes of a rational function.**

RELATED SKILLS

- I — Determine horizontal or slant asymptotic behavior.
- I — Determine vertical asymptotes from noncanceling denominator zeros.
- I — Identify a removable discontinuity and its coordinates.
- A — Cancel common factors rather than individual terms.
- A — Factor numerators and denominators completely.

☐ **2. I can sketch and analyze a rational function using its algebraic and asymptotic features.**

RELATED SKILLS

- I — Determine the sign of a rational function on intervals.
- D — Determine horizontal or slant asymptotic behavior.
- D — Determine vertical asymptotes from noncanceling denominator zeros.
- D — Identify a removable discontinuity and its coordinates.
- A — Determine excluded values from an original denominator.

☐ **3. I can interpret limit notation as a statement about function behavior near an input.**

RELATED SKILLS

- I — Read and write limit notation with correct approach direction.

☐ **4. I can estimate a limit from a graph or table.**

RELATED SKILLS

- I — Estimate a limit from input-output values approaching from both sides.
- I — Estimate one-sided and two-sided limits from a graph.

- [ ] **5. I can evaluate a limit algebraically using direct substitution or simplification.**

RELATED SKILLS

- I Resolve an indeterminate algebraic form by factoring or rationalizing.
- I Test a limit using direct substitution.
- A Factor numerators and denominators completely.

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- [ ] **6. I can interpret an infinite limit as vertical-asymptotic behavior.**

RELATED SKILLS

- I Interpret unbounded output near a finite input as an infinite limit.
- D Determine vertical asymptotes from noncanceling denominator zeros.

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- [ ] **7. I can use limit notation to describe behavior at infinity.**

RELATED SKILLS

- I Compare dominant terms to determine behavior at infinity.
- D Determine horizontal or slant asymptotic behavior.

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- [ ] **8. I can analyze the long-run behavior of quotients involving polynomial, exponential, or logarithmic expressions.**

RELATED SKILLS

- D Compare the long-run growth or decay of functions from different families.
- A Compare dominant terms to determine behavior at infinity.
- A Compare functions shown in different representations.

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- [ ] **9. I can compare the long-run growth of polynomial, exponential, logarithmic, and rational functions.**

RELATED SKILLS

- I Apply the long-run growth hierarchy among logarithmic, polynomial, and exponential functions.
- A Compare functions shown in different representations.

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- [ ] **10. I can build or select a rational or rational-adjacent model for a context and interpret its long-run behavior.**

RELATED SKILLS

- D Explain a model parameter in the language of its context.
- D Select a function family whose behavior fits a table, graph, or context.
- A Determine horizontal or slant asymptotic behavior.
- A Restrict a model to meaningful contextual inputs.